



ASSESSMENTS OF SOCIO-ECONOMIC FACTORS INFLUENCING SMALLHOLDER VEGETABLE-FARMERS' INCOME IN SOME SELECTED LOCAL GOVERNMENT AREAS OF KADUNA STATE, NIGERIA

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ABSTRACT

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The study assessed socioeconomic factors influencing the income of smallholder vegetable farmers in selected Local Government Areas of Kaduna State, Nigeria. A multistage sampling technique was used to select farmers for this study. The first stage involve purposive selection of two (2) Local Government Areas. The second stage involve a random selection of two villages from each Local Government Areas viz; UnguwarFatika, Samaru, Tudun Wada and Bomo. Finally, one hundred (100) vegetable farmers were randomly selected using balloting. Data were elicited through structured questionnaire and personal interview schedule. Descriptive statistics were employed to describe the socioeconomic characteristics of smallholder vegetable farmers. Results revealed that the mean age of the farmers are 36 years inferring an active working age bracket. The mean for household size stood at 5 persons contributing to family labour. Majority (57%) of farmers sourced their capital through personal savings indicating the need for alternative funding. The Logit regression model was used to assess the effect of vegetable production on the income of farmers in the study area. The results show that age, credit and years of experience are the determinant of income which is statistically significant, age and credit had negative coefficients while years of experience had positive coefficients. The coefficient of age of the household (-0.0763) is negative and significant at 10% indicating reasonable strong relationship between age and income. Credit access has a negative effect on income implies that the more farming household have access to credit the poorer they tend to become as most of them divert the loan for other purposes. It was therefore, recommended that key stakeholders in vegetable industry comprising the government and non-governmental organizations should assist farmers in subsidizing inputs to increase their income.

INTRODUCTION

INTRODUCTION

Vegetables are of great nutritional value in Nigeria and other sub-Saharan Africa and majorly comprise of minerals, vitamins, proteins, carbohydrates and dietary fibres. They form an important component in the human diet (Aju and Popoola, 2010). Vegetable enterprise play great economic role in the agricultural sector as the value chain can provide income and reduces poverty and unemployment to different actors of the food value chain viz; input suppliers, farmers, traders, transporters, processors, and other supportive line agencies. Vegetable farming is an important sub-sector to increase income, reduce poverty, and improve nutrition, particularly in the developing countries. Nigeria economy has long

been relying on agriculture, which shares more than 30% of the gross domestic product (GDP) where vegetable is one of the major components in contributing to the Agriculture Gross Domestic Product (AGDP) National Bureau of statistics (NBS), 2021. Despite the huge contribution to the AGDP, the growth rate of this sector has decreased and the productivity remains low (Stoeva, 2012).

The industrial potential of many vegetables available in Nigeria is enormous. What Nigerians need to do is embark on massive production of these vegetables not only for their high nutritive value but for enhancing the establishment of many processing industries (Erinle, 2019). The development of their industrial uses will stimulate large scale production of the crops and enhanced diversification of entrepreneur to site processing plants in the rural areas which will improve the quality of life of the rural population and reduce the rate of rural-urban drift.

Vegetables generate more jobs per hectare, on-farm and off-farm, than staple based agricultural enterprises (Ali *et al.*, 2012). This benefits farmers and landless laborers in both rural and urban areas. Value addition to vegetables generates further employment in the associated agri-businesses and further down the commodity chain from the producer to the consumer. Vegetables can generate higher profits than staple crops, especially when land is relatively scarce and labor is abundant (Cock and Voss, 2004; Gabre- Madhin and Hagglade, 2003; Minot and Ngigi, 2004; Subramanian *et al.*, 2010).

Furthermore, the extent to which inputs supply and productivity relates to a shortfall in food demand and supply in Nigeria is a source of concern. However, it is pertinent to note that the productivity of smallholder farmers and its contribution to the economy, food security and poverty reduction depend on the services provided by well-functioning ecosystems, including soil fertility, freshwater delivery, pollination and pest control. Smallholder farming practices, in turn, affect the condition of ecosystems. These impacts are not always negative, but poverty and immediate needs can drive smallholders to put pressure on ecosystems and this is worrisome. Despite the nutritional need of Nigerians, the daily consumption of vegetables is insufficient in Nigeria and this calls for attention (Chubike, 2013). Therefore this study become imperative since effective ecosystem can increase vegetable output. Therefore, study assessed the socio-economic characteristics influencing vegetable production of smallholder farmers in selected Local Government Areas of Kaduna state, Nigeria.

Finally, the result of the study will be useful to policy makers as a guide in designing appropriate policies needed to improve the efficiency of vegetable production and accordingly generate income to the farmers and other chain actors.

METHODOLOGY

Study Area

Kaduna State, Nigeria. It is located between Latitude 9° 03' and 11° 32' North of the Equator and Longitudes 6° 05' and 8° 38' East of the Greenwich Meridian (Kaduna State Agricultural Development Agency KADA, 2007; Kaduna State Government, KSG, 2015). It share common boundaries with Kano and Kastina to the north, Bauchi and Plateau State to the North East, Federal Capital Territory Abuja and Nasarawa State to the South, Niger and Zamfara State to the South West (Alabi and Abdulazeez, 2018; Yakubu, 2015).

It has a total land mass of 46,053 square kilometers (KSG, 2015) with a population of 6,113,503 people as at 2006. The projected population of the State is put at 9,476,053 people in 2020 at a growth rate of 3.18% per annum. The climate of the state is the tropical dry-and-wet type, classified by Koppen's. The wet season lasts from April through mid-October with a peak in August, while the dry season extends from mid- October of one

calendar-year to April of the next (Abajeet *al*, 2010). The annual average rainfall in the state is about 1323mm. The spatial and temporal distribution of the rain varies, decreasing from an average of about 1733mm in Kafanchan-Kagoro areas in the South of the study area to about 1203mm in the central part (Kaduna) and about 1032mm in Zaria, Ikara and Makarfi LGAs in the north.

The highest average air temperature normally occurs in April (28.90C) and the lowest in December (22.90C) through January (23.10C). The mean atmospheric relative humidity ranges between 70-90% and 25-30% for the rainy and dry seasons respectively. Kaduna State consists of four (4) Agricultural zone viz; BirninGwari, Lere, Maigana and Samaru-Kataf. The state consist of 23 local government area; BirniGwari, Chikun, Igabi, Ikara, Jaba, Jema'a, Kachia, Kudan, Kajuru, Kaura, Kauru, Kubua, Kaduna North, Kaduna South, Lere, Markafi, Giwa, Kagarko, SabonGari, Sanga, Soba, Zangon-kataf and Zaria.

Kaduna State has different ethnic groups including Hausa, Fulani, Gbagyi, Karama, Bajju, Kataf, Kadara and Jaba and so on including other ethnic groups from various parts of the country. The major religions are Islam and Christianity, followed by Traditional African Religion as minority. In addition to vegetable, maize, cowpea, sorghum, soybeans, potato, ginger, groundnut and other crops are grown, while other farmer rear animal including goat, sheep, cow, pigs, vegetable, duck and their likes. The predominant occupations in the area are farming, complimented by other livelihood (off-farm) activities such as trade, collection and gathering of firewood's, fruit and vegetable. Majority of the farmers are small scale farmers who cultivate vegetable alongside other crops alongside other crops for household consumption, while part of their produce including cash crop harvested is sold for income generation.

Sampling Techniques

A multistage sampling technique was used to select farmers for this study. At the first stage, a purposively selection of 2 Local Government Area, due to preponderance of vegetable production. At the second stage, a random selection of two villages from each Local Government Area which are UnguwarFatika ,Samaru, Tudun Wada and Bomo. Finally, one hundred (100) vegetable farmers were randomly selected using balloting.

Data Analytical Techniques:

Descriptive statistics

Descriptive statistics was used to describe the socio-economic characteristics involved in the production of vegetable in the study area. Arithmetic means, frequency distribution, percentages was used in cross- tabulation.

Logit Regression Model

Logit regression model was used to assess the effect of vegetable production on income of farmers. A logit regression model is a binary choice model in which a dichotomous response variable is considered as the dependent variable. (Pindyck and Rubinfeld, 1991). The dependent variable take a value of the effect of vegetable production on income of farmers .The model is expressed as follows;

$$\begin{aligned} \ln \{P_i / (1 - P_i)\} \\ = a_0 + a_1X_{1i} + a_2X_{2i} + a_3X_{3i} + \dots + a_9X_{9i} \\ + U \dots \dots \dots (2) \end{aligned}$$

Where the subscript i denote the ith in the sample, P is the probability of the outcome, a₀ is the intercept or constant term and a₁-----a₆ are the coefficient associated with each of explanatory variable, X₁, X₂, X₃ ---and U is the error term. The likelihood of observing the

samples is formed by introducing a dichotomous dependent. The factor hypothesized to determine the income of the 1th household are defined as follows

X_1 = Age of farmer (years)

X_2 = Marital Status (Single=0, married=1, widowed=2, divorced=3)

X_3 = Level of education of farmer (years of schooling)

X_4 = farm size (ha)

X_5 = Household size (Number of members)

X_6 = Farming experience (years)

X_7 = Access to credit (Access=1, no access=2)

X_8 = Membership of co-operative society (years)

X_9 = Output (kg)

RESULT AND DISCUSSION

Socioeconomic characteristics of vegetable farmers.

The socioeconomic characteristics of the vegetable farmers do greatly influence the behavior of vegetable farmers in the study area. Some of these characteristics are age, marital status, household size, educational background, access to credit, farm size and years of experience.

Distribution of respondents by age

The distribution of the farmers according to age showed that about 41% of the farmers were within the active age of 20-30 years while 32% of them are 31-40 years of age with the mean age of 36.07 years. This implies that these farmers were within the active working age bracket. These farmers were particularly young people who could afford to venture into the vegetable business which is known to be characterized by risk such as disease, drought and theft and they are matured to take credit decision that can sustain their vegetable production. This finding agrees with the findings of Sekumade and Toluwase, (2014).

Household size

The result of the analysis on household size showed that about 60% of the farmers had household size of 1-5 persons, while the least was 6-10 persons households (35%) with the mean household size of 5 persons. Higher number of members of the household contributes to family labour. This supports most of the studies that confirmed large house size among the farming households where they see family size as a work force that supply the most needed labour requirement for production activities in the study area (Emaikwuet *al.*, 2011).

Farming Experience of Respondent.

The findings in table 1 revealed that 54% of the farmers had 1-5 years of experience, 30% had 6-10 years, 8 % had 11-15 years, and while 8% of the farmers had more than 20 years of experience. The mean years of experience was found to be 7 years. This suggests that majority of the vegetable farmers in the study were new entrants into the business. It is generally expected that productivity increases with years of experienced (Umar, M. 2013).

Farmers master the techniques of production and avoid previous mistakes. Experienced vegetable farmers are likely to make better decisions to enhance productivity and income, because it is expected that experience in vegetable production usually determines the effectiveness of farmers' decision with respect to inputs combinations or resource allocation. Years of farming experience affects the productivity and technical efficiency of vegetable production. The longer the years of farming experience, the more the knowledge acquired, the more efficient the farmer become.

Membership of Cooperative.

The result of the distribution of farmers showed that 63% were non-members of cooperative societies, while only about 37% of the farmers were members of cooperative societies. This implies that the benefit of cooperative membership was not fully accessed by the farmers. Cooperative membership avail co-operators the benefit of pooling resources together. Members also benefit from enhanced delivery of agricultural extension services, especially to farm financial institutions, government and private institutions.

Source of Credit.

The distribution of farmers according to their source of credit is presented in Table 1, it shows that 57.0% sourced for credits through personal savings that possibly might have been accumulated over time and from sale of their products, and this is closely followed by friend and family as asserted 31.0%, while 12.0 % sourced from commercial banks. Access to credit is one of the major determinant of outputs and effective productivity of farmers.

Table 1: Socioeconomic characteristics of yam marketers.

Age Range	Frequency	Percentage (%)
0 < 30	41	41
31 – 40	32	32
41 – 50	16	16
51 – 60	7	7
>61	4	4
Total	100	100
Mean age	36.07	
Std Dev.	11.03	
Minimum	23	
Maximum	65	
Household size	Frequency	Percentage
1 – 5	60	60
6 -10	35	35
11 – 15	5	5
Total	100	100
Mean	4.95	
Std. Dev	3.31	
Minimum	1	
Maximum	15	
Years of Exp	Frequency	Percentage
1-5	54	54
6-10	30	30
11-15	8	8

16-20	6	6
21-25	2	2
Total	100	100
Cooperatives		
Membership	37	37
Non-Membership	63	63
Total	100	100
Source of capital		
Personal saving	57	57
Friends and family	31	31
Commercial banks	12	12
Total	100	100

Source: Computed from field survey data, 2021.

Assessmnt of Vegetable Production on Income of Farmers.

The Logit regression model was used to assess the effect of vegetable production on the income of farmers in the study area. Table 4.30 shows the coefficients of the Logit model with other estimated parameters. The likelihood ratio test was 53.113 with 7 degree of freedom significant at ($p \leq 0.01$). This indicates a joint significant of all the variables in influencing the income of vegetable farmers which means that the socioeconomic characteristics of the farming households have significant effect on their income. The results in table 2 show that age, credit and years of experience are the determinant of income which is statistically significant age and credit had negative coefficients while years of experience had positive coefficients.

The coefficient of age of the household (-0.0763) is negative and significant at 10% indicating reasonable strong relationship between age and income. This suggests that the older the household head the higher the mean household consumption which is brought about by a reduction in mean household income. This can be attributed to the fact that with increasing age comes increasing experience, so the farmer rely on experience rather than adopting new technology (Eshetu and Mekonnen, 2016), hereby reducing the ability of the household head to earn more income.

The coefficient for years of experience (0.1104) is positive and significant at the 5% level of probability. The positive coefficient indicates direct linkage between experience and income. This implies that farmers will experience increased output given the number of years they have spent farming, they are expected to have gained enough knowledge, sometimes count more on their experiences than educational attainment in order to increase their productivity which consequently leads to higher return on investment.

The coefficient for marital status (-0.6939) was found to be negative and statistically significant at the 10% level of probability. The negative coefficient indicates an indirect linkage between marital status and income of the farmers. The possible reason could be as marital status tends to increase the numbers in the family, it means the more food and other necessities are needed and thus the demand for income rises.

Credit access has a negative effect on income implies that the more farming household have access to credit the poorer they tend to become. Access to credit is significant at 10%. This agrees with the findings of Nse-Nelson *et al* (2016), that funds obtained from loans are used for non-profitable purpose example household consumption rather than productive purpose. At the point of loan repayment, especially at high interest rate, the farming

household disposable income decreases hence the decreased ability to afford optimum consumption levels.

Table 1: Estimate of the effect of vegetable production on the income of farmers in the study area

Variables	Coefficient	Standard error	T-ratio
Constant	4.6484	1.5116	3.08
Age	-0.0763*	0.029	-2.63
Marital status	-0.6939*	0.5355	-1.30
Education	-0.2088	0.1715	-1.22
Household size	-0.1051	0.0977	-1.08
Years of Experience	0.1104**	0.0558	1.98
Credit	-0.6775*	0.52	-1.30
Number of observation	100		
LR chi2(6)	25.57		
Prob> chi2	0.0003		
Pseudo R2	0.194		
Log likelihood ratio test	53.113		

Note Source: computed from field survey data, 2021.

Note: ***, ** and * imply significant at 0.01, 0.05 and 0.1 levels, Ns implies not significant. Values in parentheses are the respective t – ratios

Constraints Faced in Vegetable Production in the Study Area.

About 76% of the farmers ranked incidence of Pest and Disease attack as the major constraint in vegetable production. Vegetable requires prompt application of agrochemicals such as insecticides and herbicides to check the menace of pests and diseases infestation that may occur as a result of overgrowth of weeds.

About 33% ranked poor seed quality as the last constraint. Poor seed quality artificially lowered domestic vegetable levels through massive production of vegetable resulting in low vegetable of locally produced vegetable, distribution and its marketing by various stakeholders and government agencies with non-transfer of actual costs to consumers has adverse effect on vegetable production.

About 45% of the farmers ranked shortage and high cost of farming, fertilizer and inputs as constraint. This is one of the problems confronting vegetable production in the study area. Farmers in the study area are poor and cannot afford to buy some of the farming inputs. This implies the fact that inputs are too expensive for an ordinary farmer in the study area considering farmer's financial status.

Table 2: Constraints in vegetable production in the study

S/N	Constraint of Vegetable Production	Frequency	Percentage	Ranking
1	Incidence of Pest and Disease	76	76	1 st
2	Shortage and High Cost of Fertilizer	76	76	1 st
3	Unpredictable Sales and Perishability	57	57	2 nd

4	Lack of Storage Facilities	54	54	3 rd
5	Environmental Problems	45	45	4 th
6	Shortage and High Cost of Labour	45	45	4 th
7	Poor seed quality	33	33	5 th
8	Difficulty in Accessing	27	27	6 th
9	Difficulty in acquiring farm land	27	27	6 th

Field survey: 2021

CONCLUSION AND RECOMMENDATIONS

The study establishes that vegetable farmers in the study area though in small scale has potential to increase their income generation while factors such as age, farming experience and credit respectively are determinant of vegetable production and increase in income of the farmers in the face of constraints such as pest and disease attack, high cost of farming inputs, unpredictable sales/marketing, lack of storage facilities.

The study recommend provision of competent extension agents to visit and enlighten vegetables farmers on the technicality of production processes and how they can improve using advance technology. This will not only enhance the skills of the farmers but also ensures food security and increase in income. Key stakeholders in vegetable industry comprising the government and non-governmental organizations could assist farmers to increase their use of improved seeds by supporting them with production capital with which they can purchase the inputs and also provide machineries to reduce the cost of labor by subsidizing the vegetables of the working hour thereby enhancing their income generation. This is because use of these inputs in vegetable production enhances the outputs which translate to higher returns. Farmers should be encouraged to form cooperatives to enable them benefit from such cooperatives in terms of training, information on innovations, and access to credits to boost productivity and improve income.

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